

The Demand for Sex: Evidence from the United States

Team #8

April 11, 2026

Abstract

In this paper, we contribute to the robust literature on parental child sex preference, and in particular the preference for male children. We propose that the preference for male children be decomposed into (1) a bias towards sons, and (2) a parental bias towards diversity in the sexes of their children. Using microdata from the Panel Study of Income Dynamics (PSID), we find that parents whose first and second children are of the same sex are 5.3% more likely to have a third child than parents whose first and second children are of different sexes; this pro-diversity bias explains around two-thirds of the observed preference towards male children. We analyze the change in this preference over time and identify heterogeneous effects by relative intra-couple income in line with a bargaining model of fertility.

1 Introduction

Biological organisms are assigned sex randomly (Orzack et al., 2015). For this reason, humans are constrained in their ability to influence the sex of their children. However, the sex of a child is of importance to parents. Historically and across many cultures, parents have preferred male children to female children (Le and Nguyen, 2022). This fact has been attributed to differentials in the returns from male and female children (e.g. labor, dowry obligations, and elder care), differential preferences and bargaining power within couples, and sex-based prejudice (Das Gupta et al., 2003; Kumar and Sinha, 2020). As the difference in the pecuniary returns from male and female children has declined, especially in wealthy nations like the United States, parents' stated preference for male children has declined as well (Miller, 2018).

In this paper, we leverage random sex assignment of children to estimate parental preference for sex and sex diversity in the United States from 1970-2010 at an individual level. In so doing, we enter an extensive literature regarding fertility and the effects of the genders of previous children on future outcomes. Dahl and Moretti (2008) find that women with first-born daughters are less likely to marry, and parents with first-born girls are more likely to get divorced. Jones et al. (2023) examine trends in the United States of gender preferences from 1850 to 2019, finding that women are more likely to have a third child if the first two children are the same gender, and uncover evidence of slight parental preferences for males. We verify and extend these findings through a variety of means. First, we find a significant sex diversity preference among individuals with 2 children, and like Jones et al. (2023) find marginal evidence of preference for males. We extend our analytic framework to families with 3, 4, and 5+ children of the same gender, finding suggestive evidence

that sex-diversity preference is increasing in parity. We further find that diversity preference is higher when previous parity is of the opposite sex of the higher earning parent, indicating that the preference is mediated by household fertility bargaining.

2 Data

All our data was taken from the Panel Study of Income Dynamics (henceforth, PSID) constructed by the University of Michigan’s [Social Research Center](#). We draw on the Cross-Year Individual file, the Family files, and the Childbirth and Adoption History (CAH) file.¹ This panel provides information on a family’s total number of children across time, the order in which their children were born, and the sex of all children, in addition to a number of covariates including income, education status, and race of the parents; we use these covariates to perform our heterogeneity analysis.

It should be noted that even though the PSID is a rich source of panel data, our analysis relies predominantly on pooled regressions on various subsets of the sample. Consequently, we left the panel unbalanced and only excluded those observations for which one or more of the variables of interest was unobserved.

2.1 Sample Definition

The sample was restricted to the collection of all households observed in the PSID, starting 1968, that had at least one non-adopted child born between 1970 and 2010; all households observed had to have two household heads at the time the first child was born. This gives us a total of $n = 7187$ unit-time observations, after excluding households for which the PSID weights of either household head were absent.

For the purposes of our analysis, it is also useful to further subset the data to households with at least k non-adopted children born between 1970 and 2010, reducing the sample size to $n = 5270$ in the $k = 2$ case. To conduct a heterogeneity analysis, divide our sample based on the relationship between the incomes of the two heads (see Section 4.3).

3 Empirical Strategy & Identification

We first define the variable C_i to be the total number of children of household i by the end of their child-bearing period. The gender of each child $c_{i,j} \in \{M, F\}$ for household i and child number j , and male and female respectively. Consider this expositional regression specification, which applies *only* to households where $C_i > 0$:

$$\mathbb{1}\{C_i > 1\} = \beta_0^{(1)} + \beta_1^{(1)} \mathbb{1}\{c_{i,1} = F\} + \varepsilon_i^{(1)} \quad (1)$$

This regression measures the additional probability of having more children after first having a female child, than first having a male one. In other words, it measures the degree to which households are unfulfilled by having a female child, and opt to have another child in hopes of having

¹All observations are weighted using the PSID’s Individual weights.

a male child. In mathematical terms:

$$\beta_1^{(1)} = \mathbb{P}[C_i > 1 \mid c_{i,1} = F] - \mathbb{P}[C_i > 1 \mid c_{i,1} = M]$$

Now consider this specification, applying to households where the first child was a girl:

$$\mathbb{1}\{C_i > 2\} = \beta_0^{(2)} + \beta_1^{(2)} \mathbb{1}\{c_{i,1} = c_{i,2} = F\} + \varepsilon_i^{(2)} \quad (2)$$

The coefficient $\beta_1^{(2)}$ provides an estimate of the additional probability of having more children, given that the first two children were female. This is a further estimate on the extent to which households want male children over female children. Given that my first child was female, how much more likely am I to have a third child if my second child was female? Mathematically,

$$\beta_1^{(2)} = \mathbb{P}[C_i > 2 \mid c_{i,1} = F, c_{i,2} = F] - \mathbb{P}[C_i > 2 \mid c_{i,1} = F, c_{i,2} = M]$$

We can do the same for male children:

$$\mathbb{1}\{C_i > 2\} = \beta_0^{(3)} + \beta_1^{(3)} \mathbb{1}\{c_{i,1} = c_{i,2} = M\} + \varepsilon_i^{(3)} \quad (3)$$

And we get our expression for $\beta_1^{(3)}$:

$$\beta_1^{(3)} = \mathbb{P}[C_i > 2 \mid c_{i,1} = M, c_{i,2} = M] - \mathbb{P}[C_i > 2 \mid c_{i,1} = M, c_{i,2} = F]$$

We now can derive a nice interpretation for the difference in the two coefficients $\beta_1^{(2)} - \beta_1^{(3)}$:

$$\begin{aligned} \beta_1^{(2)} - \beta_1^{(3)} &= (\mathbb{P}[C_i > 2 \mid c_{i,1} = F, c_{i,2} = F] - \mathbb{P}[C_i > 2 \mid c_{i,1} = F, c_{i,2} = M]) \\ &\quad - (\mathbb{P}[C_i > 2 \mid c_{i,1} = M, c_{i,2} = M] - \mathbb{P}[C_i > 2 \mid c_{i,1} = M, c_{i,2} = F]) \\ &= (\mathbb{P}[C_i > 2 \mid c_{i,1} = F, c_{i,2} = F] + \mathbb{P}[C_i > 2 \mid c_{i,1} = M, c_{i,2} = F]) \\ &\quad - (\mathbb{P}[C_i > 2 \mid c_{i,1} = F, c_{i,2} = M] + \mathbb{P}[C_i > 2 \mid c_{i,1} = M, c_{i,2} = M]) \end{aligned}$$

By the Law of Total Probability, we can expand the following two expressions:

$$\begin{aligned} \mathbb{P}[C_i > 2 \mid c_{i,2} = F] &= \mathbb{P}[C_i > 2 \mid c_{i,1} = F, c_{i,2} = F] \cdot \mathbb{P}[c_{i,1} = F] \\ &\quad + \mathbb{P}[C_i > 2 \mid c_{i,1} = M, c_{i,2} = F] \cdot \mathbb{P}[c_{i,1} = M] \\ &= \frac{1}{2} (\mathbb{P}[C_i > 2 \mid c_{i,1} = F, c_{i,2} = F] + \mathbb{P}[C_i > 2 \mid c_{i,1} = M, c_{i,2} = F]) \\ \mathbb{P}[C_i > 2 \mid c_{i,2} = M] &= \mathbb{P}[C_i > 2 \mid c_{i,1} = F, c_{i,2} = M] \cdot \mathbb{P}[c_{i,1} = F] \\ &\quad + \mathbb{P}[C_i > 2 \mid c_{i,1} = M, c_{i,2} = M] \cdot \mathbb{P}[c_{i,1} = M] \\ &= \frac{1}{2} (\mathbb{P}[C_i > 2 \mid c_{i,1} = F, c_{i,2} = M] + \mathbb{P}[C_i > 2 \mid c_{i,1} = M, c_{i,2} = M]) \end{aligned}$$

Now we can rewrite the difference $\beta_1^{(2)} - \beta_1^{(3)}$ as the following:

$$\begin{aligned}
\beta_1^{(2)} - \beta_1^{(3)} &= (\mathbb{P}[C_i > 2 \mid c_{i,1} = F, c_{i,2} = F] + \mathbb{P}[C_i > 2 \mid c_{i,1} = M, c_{i,2} = F]) \\
&\quad - (\mathbb{P}[C_i > 2 \mid c_{i,1} = F, c_{i,2} = M] + \mathbb{P}[C_i > 2 \mid c_{i,1} = M, c_{i,2} = M]) \\
&= 2\mathbb{P}[C_i > 2 \mid c_{i,2} = F] - 2\mathbb{P}[C_i > 2 \mid c_{i,2} = M] \\
&= \boxed{2(\mathbb{P}[C_i > 2 \mid c_{i,2} = F] - \mathbb{P}[C_i > 2 \mid c_{i,2} = M])} \tag{*}
\end{aligned}$$

The interpretation of the difference in the coefficients is now intuitive: $\beta_1^{(2)} - \beta_1^{(3)}$ is proportional to the difference between the excess probability of having an additional child given that the second child is female versus male, *unconditional* on the gender of the first child. It is a pure representation of male/female bias, with no interpretation of diversity preference, as it cancels out. An alternate explanation is this: The first term above Equation (*) includes bias for males and bias for diversity. The fourth term includes bias for females and bias for diversity. The two bias for diversity terms cancel out as well. The second and third terms are agnostic to bias towards males or females, and instead includes only bias for diversity, which cancels. We are then just left with the difference between bias for males versus females.

We can now generalize this specification for arbitrary k , where k denotes various numbers of children. This regression only applies to households where $c_{i,1} = \dots c_{i,k-1}$:

$$\mathbb{1}\{C_i > k\} = \beta_0^{(4)} + \beta_1^{(4)} \mathbb{1}\{c_{i,k} = c_{i,k-1}\} + \varepsilon_i^{(4)} \tag{4}$$

The interpretation of this result is different. Rather than estimating the preferences for males versus females, this estimates how much households prefer diversity. If my first $k - 1$ children were the same gender, and I have another child of that same gender, am I more likely to have even more children to chase gender diversity in my household? We can write $\beta_1^{(4)}$ in the following way:

$$\beta_1^{(4)} = \mathbb{P}[C_i > k \mid c_{i,1} = \dots = c_{i,k-1} = c_{i,k}] - \mathbb{P}[C_i > k \mid c_{i,1} = \dots = c_{i,k-1} \neq c_{i,k}]$$

4 Results

4.1 Decomposing Sex Preferences

The OLS estimate of $\beta_1^{(1)}$ in Equation 1 is 0.008, and statistically insignificant. Intuitively, this means that a couple's first child being female does not cause them to be more likely to have another child, providing no evidence for sex preference.

Running the generalized specifications in Equations 2 and 3, we obtain a highly statistically significant value of $\hat{\beta}_1^{(2)} = 0.080$; $\hat{\beta}_1^{(3)} = 0.028$ has the same standard error and is not significant at the 5% level. However, when we run the regression in Equation 4 with $k = 2$ (which is agnostic to sex), we obtain a coefficient $\hat{\beta}_1^{(4)} = 0.053$, which is again highly significant.

This provides a more complete picture of sex preference; on the one hand, there is a distinct, significant preference for additional children given two female children that does not exist for two male children; this is an indicator of pro-male bias (see Section 4.2). On the other hand,

the coefficient $\hat{\beta}_1^{(4)}$ is interpretable as the pro-diversity bias that partially explains why there is a perceived preference towards male children.

To isolate that part of the pro-male preference that is unexplained by diversity bias, we can simply take the difference of the two ATEs to get 0.027. That is, around 33% of the pro-male preference is explained by pure pro-male bias; and 66% is explained by diversity bias. This helps us understand why the OLS estimate in Equation 1 has such low significance; there is no impact of diversity in that model.

To further understand demand for diversity as a function of all-same-sex parity, we plot the OLS estimates of $\beta_1^{(4)}$ from Equation 4 for $k = 2, 3, 4, 5$ (See Figure A). Aside from the deviation for $k = 4$, the OLS estimates are strictly increasing, exhibiting increasing demand for diversity as parents have more children of the same sex.

4.2 Testing for Pro-Male Bias & Pro-Diversity Bias

The interpretation of the coefficients $\beta_1^{(2)}$ and $\beta_1^{(3)}$ in Section 3 furnishes us with a useful way to test for bias towards male children by analyzing the difference $\hat{\beta}_1^{(2)} - \hat{\beta}_1^{(3)}$. A simple Wald test of this difference, with $H_0 : \beta_1^{(2)} - \beta_1^{(3)} \leq 0$, gives an estimate of 0.05228 with a standard error of 0.02752. The one-sided p -value is 0.02872, indicating statistical significance at the 5% level. Consequently, we reject the null hypothesis that $\hat{\beta}_1^{(2)} - \hat{\beta}_1^{(3)} \leq 0$ and find evidence of statistically significant pro-male bias. On the other hand, significance of diversity bias is clear from the significance of the positive estimate of $\beta_1^{(4)}$ in the $k = 2$ case.

4.3 Heterogeneity Analysis

We now examine a potential means through which demand for sex might be mediated. We run regressions specified in Equations 2 and 3 on two disjoint subsets of the sample. The first subset is comprised of households where the man earns more, and the second is comprised of households where the woman earns more. In the first subset we obtained a significant positive estimate of $\beta_1^{(2)}$, which was not the case for the second; the reverse is true for $\beta_1^{(3)}$, which was positive and significant in households where the woman earned more. Intuitively, demand for gender diversity is high when a couple's first two children are of the opposite sex of the couple's highest earning member. We relate this result to two possible mechanisms.

Firstly, in a household in which the man earns higher, the perceived returns to having a male child increase. Consequently, these households are more likely to incur the costs of having additional children until they have a boy. The converse holds in a household wherein the woman earns higher. This difference in perceived returns to having a boy versus a girl declines, and these households are more likely to try until they have a girl.

Secondly, the household head that earns more has higher bargaining power. Assuming they are likely to favor having children of the same sex as them, earning more gives them more power in household decisions and consequently, their household is more likely to continue having children until they have a child of their preferred sex.

Table 1: OLS Estimates of First-Child Sex on
Second Birth

	$\mathbb{1}\{C_i > 1\}$
$\mathbb{1}\{c_{i,1} = F\}$	0.008 (0.009)
Observations	7,187

SEs in parentheses, PSID weights.

Signif. codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table 2: OLS Estimates of Third-Birth Regressions

	$\mathbb{1}\{C_i > 2\}$		
	Two Female	Two Male	Same Sex
$\mathbb{1}\{c_{i,1} = c_{i,2} = F\}$	0.080*** (0.019)		
$\mathbb{1}\{c_{i,1} = c_{i,2} = M\}$		0.028 (0.019)	
$\mathbb{1}\{c_{i,1} = c_{i,2}\}$			0.053*** (0.014)
Observations	2,623	2,647	5,270

SEs in parentheses, PSID weights.

Signif. codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, $^\dagger p < 0.1$.

Table 3: Third-Birth Effects by Parental Earnings Pattern

	$c_{i,(1,2)} = (F, F)$ vs. $c_{i,(1,2)} = (F, M)$		$c_{i,(1,2)} = (M, M)$ vs. $c_{i,(1,2)} = (M, F)$	
	Man Earns More (1)	Woman Earns More (2)	Man Earns More (3)	Woman Earns More (4)
$\mathbb{1}\{c_{i,(1,2)} = (F, F)\}$	0.111* (0.051)	0.008 (0.039)		
$\mathbb{1}\{c_{i,(1,2)} = (M, M)\}$			0.065 (0.053)	0.112** (0.038)
Observations	378	648	363	672

Notes: Dependent variable is $\mathbb{1}\{C_i > 2\}$. Standard errors in parentheses. Regressions are unweighted. Signif. codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, $^\dagger p < 0.1$.

5 Conclusion

We find that in the American setting, parents exhibit demand for male children and demand for sex diversity. However, we cannot identify which of these demands is stronger. We find further that first child sex does not directly affect subsequent fertility behavior, and find suggestive evidence that demand for sex diversity is increasing in same-sex parity (See Appendix Figure A). We also identify interesting temporal trends in demand for male children and demand for sex diversity. In the early 1970s, parents preferred male sex homogeneity; by the 1980s, they preferred the opposite (See Appendix Figure C). Parental desire for sex diversity increased sharply from the 1980s to the 1990s (See Appendix Figure D). Finally, we find evidence that demand for sex diversity is mediated by household bargaining. When the woman in a couple earns more, the demand for sex-based diversity increases, but only when both children are male; when the man in a couple earns more, the demand for sex-based diversity increases, but only when both children are female. These findings provide insight into couples' revealed fertility preferences, how they have changed over time, and the mechanisms underlying them.

References

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Appendix

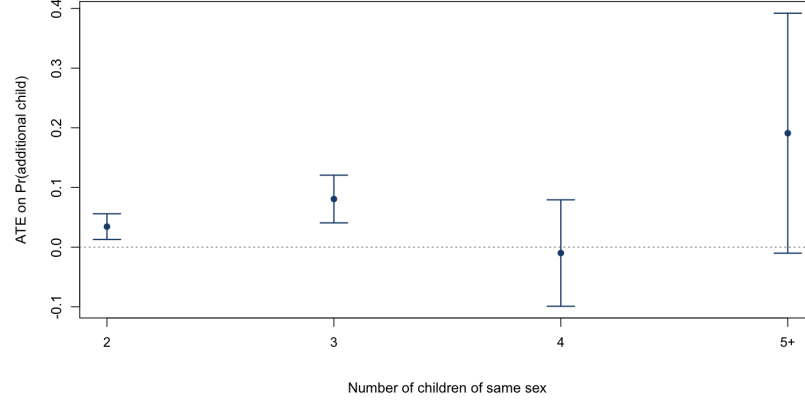


Figure A: $\beta_1^{(4)}$: Added probability of having a child after $c_{i,1} = \dots c_{i,k}$ versus $c_{i,1} = \dots c_{i,k-1} \neq c_{i,k}$. Sample size: $n = 5270$ for $k = 2$, $n = 2180$ for $k = 3$, $n = 473$ for $k = 4$, $n = 95$ for $k \geq 5$.

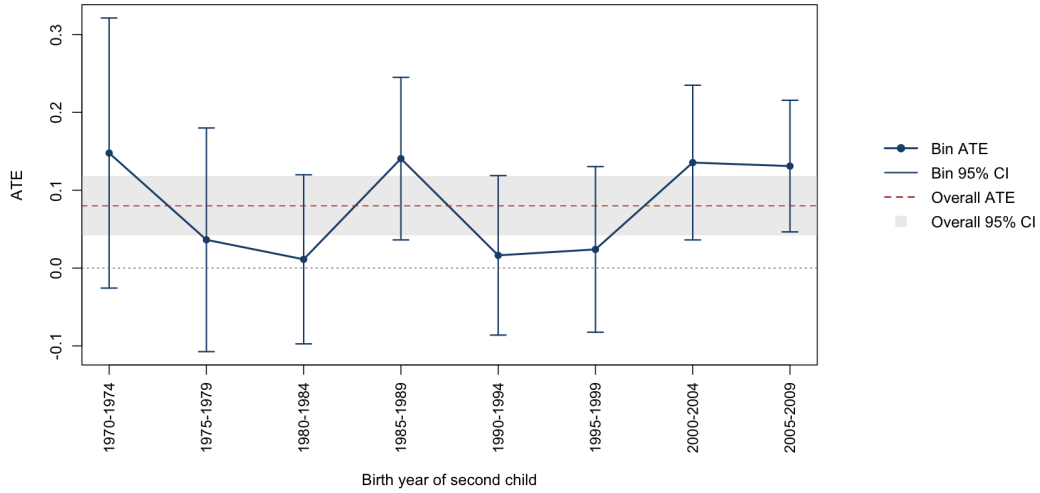


Figure B: $\beta_1^{(2)}$: Additional probability of having a third child after $c_{i,(1,2)} = (F, F)$ versus $c_{i,(1,2)} = (F, M)$ from 1970 to 2009. Sample size: $n = 2623$.

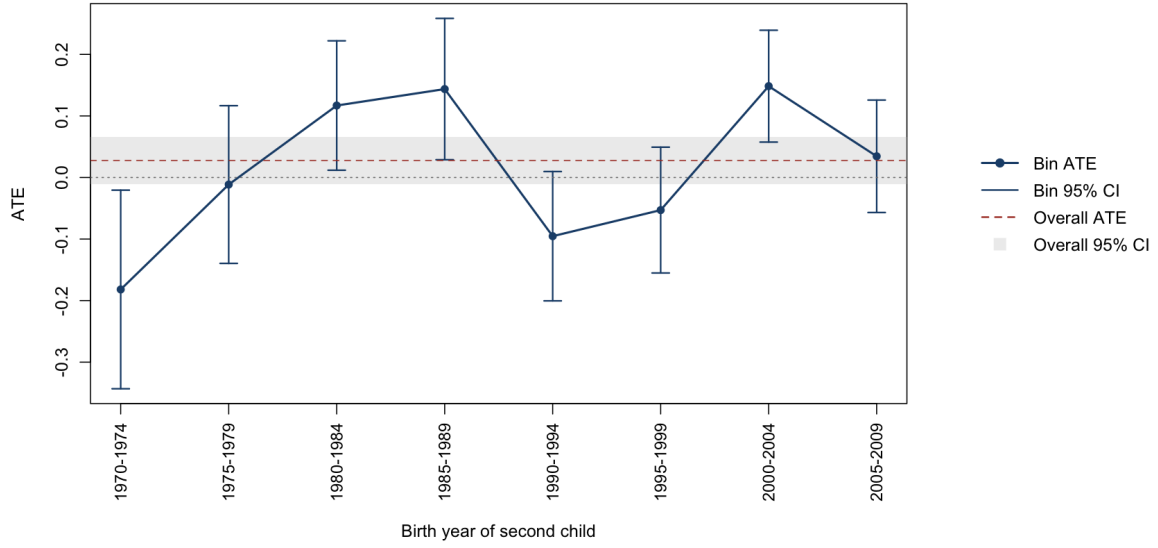


Figure C: $\beta_1^{(3)}$: Additional probability of having a third child after $c_{i,(1,2)} = (M, M)$ versus $c_{i,(1,2)} = (M, F)$ from 1970 to 2009. Sample size: $n = 2647$.

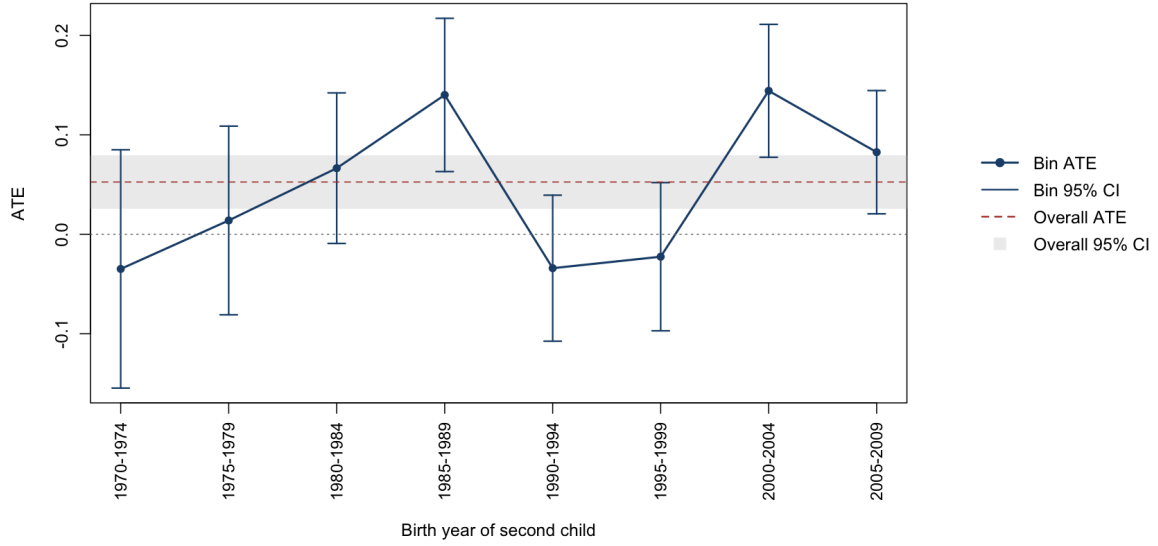


Figure D: $\beta_1^{(4)}$ ($k = 3$): Additional probability of having a child after $c_{i,1} = c_{i,2}$ versus $c_{i,1} \neq c_{i,2}$. Sample size: $n = 5270$.